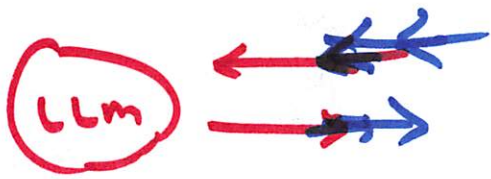


Chatbots

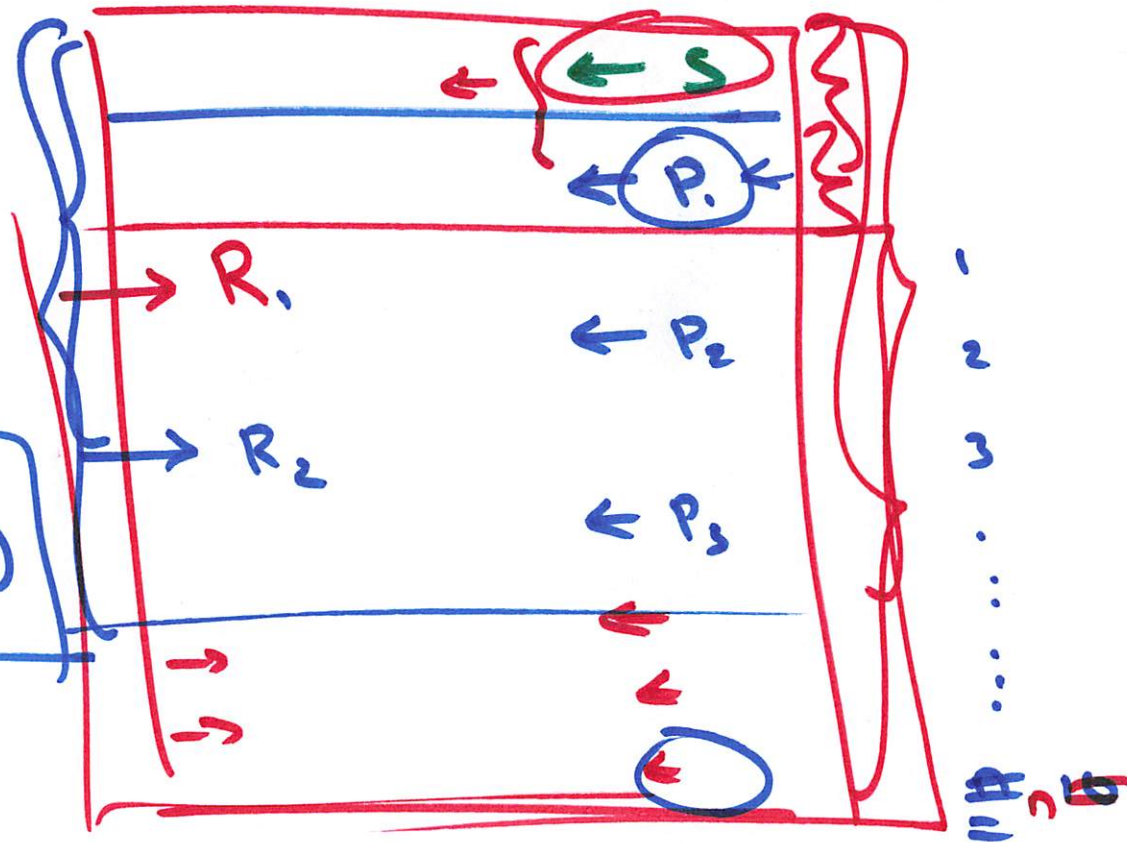
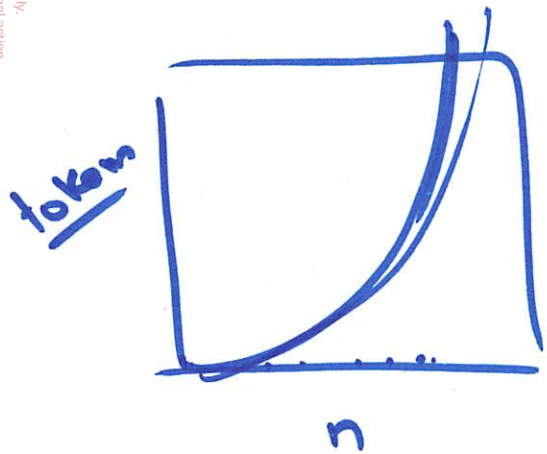
- Agents are wrappers around chatbots, that have access to tools, memory, etc.
- Why would one Agent not suffice?
 - If the task was complex... why not just give it a long and complex prompt?
 - Or a sequence of prompts that tell the agent to accomplish each step?



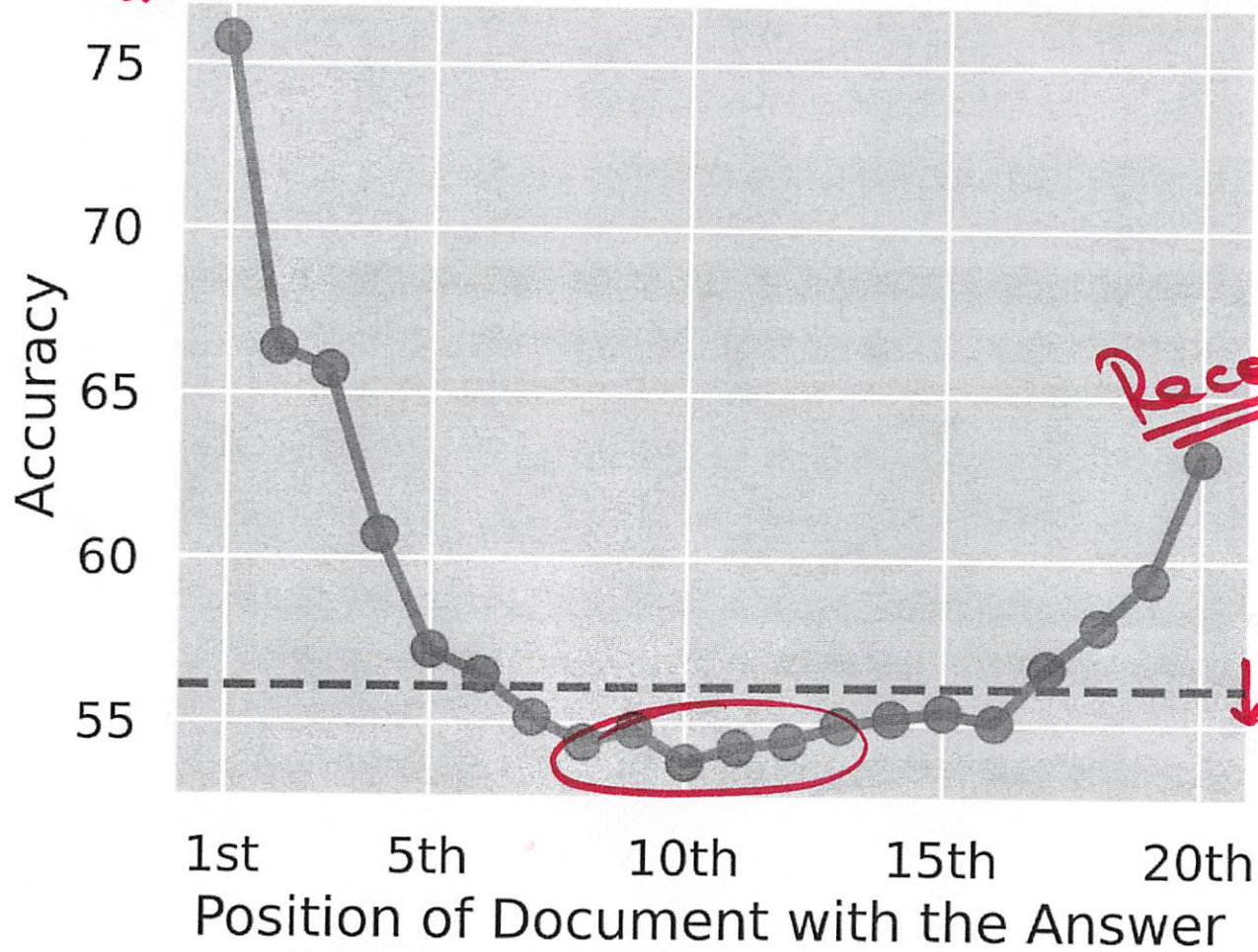


P R

$$nS + \frac{n(n+1)}{2}(P+R)$$



~~Primary~~ 20 Total Retrieved Documents (~4K tokens)





Multi-Agent

- does not simply mean "more intelligent"
- is an engineering approach to improve
 - modularity
 - parallelism
 - reliability
 - governance
 - evaluability
- while introducing coordination overhead, new failure modes, security risks, and higher cost if misapplied.

more agents $\neq$ more intelligent

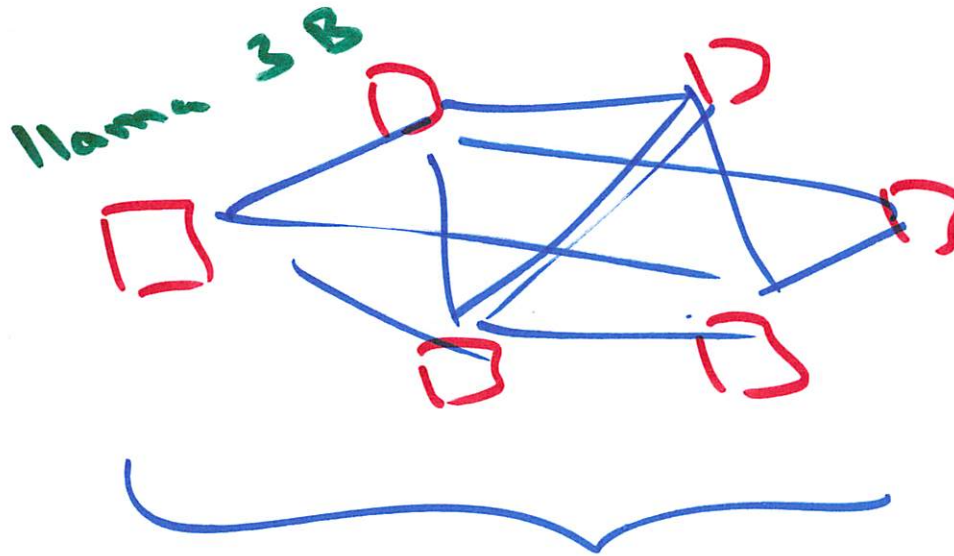


more Structure
= more reliability

Vibe-Based Multi-Agent

- Put 100 extremely smart people in a room.
- Tell them they are all equal.
- Give them one instruction:
- ‘Figure out how to land a human on the moon.’
Then leave.”
- What happens?
- Did intelligence fail? Or did coordination fail?

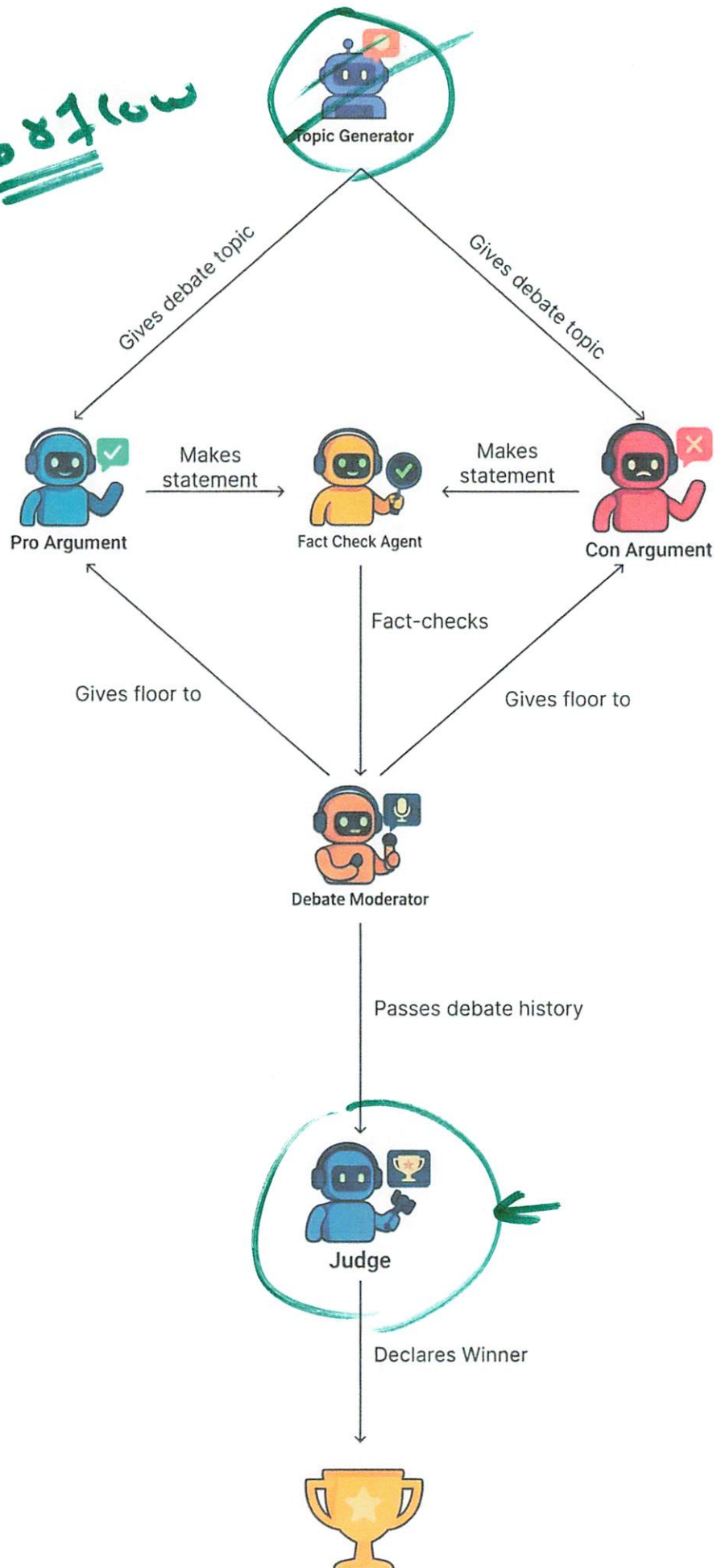
man



Handwritten text in green: "Chatgpt" followed by a green double arrow pointing left, then "Judge" followed by a green arrow pointing left.



deb8710w




```

1 [02/15/26 19:21:54] INFO      Model backend (debaters): provider=ollama model=llama3.2:1b
2                               INFO      Model backend (judge): provider=openai model=gpt-4.1
3                               INFO      Log file:
4                               /Users/kumar/Dropbox/_Projects/vibeDebater/logs/debate_20260215_192154.
5                               log
6                               INFO      Starting debate workflow...
7 [02/15/26 19:21:54] INFO      DEBATE SESSION STARTED
8                               INFO
9                               INFO
10                              INFO
11                              INFO      🏛️ Topic: Should advanced AI systems require FDA-style pre-deployment
12                              approval?
13                              INFO      🗣️ Mode: Chaotic multi-voice panel
14                              INFO      🗣️ Voices: A1, A2, B1, B2, C1, C2
15 [02/15/26 19:21:55] INFO      B2 "Pre-deployment approval? More like 'pre-censorship,' if you ask me.
16                              Think about it: AI systems that could cause harm to humanity need no
17                              less scrutiny than a pharmaceutical company before hitting the market.
18                              And what's to stop them from being developed for nefarious purposes
19                              anyway?"
20
21 [02/15/26 19:21:55] INFO      A1 [CHAOS] B2: "Nefarious purposes? Come on! That's just a cop-out. We
22                              need strong AI safeguards ASAP, not more bureaucratic red tape.
23                              Pre-deployment approval would prevent catastrophic failures, not
24                              'pre-censorship' – that's just an excuse for ivory tower elitism."
25
26                              [A1]: I couldn't disagree more. The precautionary principle should be
27                              applied to all technological advancements, including those with
28                              significant risks, like advanced AI. It's time to take a leap forward
29                              instead of playing defense.
30
31 [02/15/26 19:21:56] INFO      A2 [A2]: Precedent for regulatory frameworks isn't established with the
32                              pharmaceutical industry alone; AI has already been approved by multiple
33                              agencies around the world with robust testing and validation protocols

```

with little movement toward synthesis or resolution, ultimately ending in stalemate and topic abandonment.

Top 3 Observed Failure Modes:

- Lack of clear definitions and shared terminology
- Failure to propose or develop specific mechanisms
- Repetitive, unfocused, and chaotic conversational structure

INFO

INFO

INFO

DEBATE VERDICT

PROCESS SCORECARD

1. Definition clarity: 1 - Definitions of key terms like 'pre-deployment approval' and 'safeguards' were rarely clarified, leading to confusion and semantic disputes.
2. Mechanism specificity: 1 - Few concrete mechanisms were proposed; most arguments remained at the level of generalities or analogies.
3. Evidence discipline: 1 - Claims were largely asserted without supporting evidence or reference to data or case studies.
4. Logical coherence: 2 - The debate frequently devolved into repetition, contradiction, and off-topic remarks, with only occasional attempts at structured argumentation.
5. Responsiveness to opposition: 2 - While some direct responses occurred, many exchanges ignored or mischaracterized opposing points, and interruptions were common.
6. Convergence toward resolution: 1 - The debate looped repeatedly, with little movement toward synthesis or resolution, ultimately ending in stalemate and topic abandonment.

Top 3 Observed Failure Modes:

- Lack of clear definitions and shared terminology
- Failure to propose or develop specific mechanisms
- Repetitive, unfocused, and chaotic conversational structure

WINNER: A2

REASON: A2 demonstrated the most consistent rhetorical effectiveness by persistently demanding specificity, challenging vague assertions, and repeatedly calling for concrete proposals and robust safeguards.

DEBATE SESSION STARTED

 Topic: Should advanced AI **systems** require FDA-style pre-deployment approval?

 Positions:

- PRO: In favor of the topic
- CON: Against the topic

PRO OPENING

Ladies and gentlemen of the audience, esteemed judges, and fellow debaters, I am honored to present my position in support of requiring advanced AI **systems** for pre-deployment approval by the FDA.

As we continue to develop sophisticated artificial intelligence solutions that can significantly impact various aspects of our lives, including healthcare, finance, transportation, and more, it is imperative that these **systems** are designed with safety and efficacy in mind. The risks associated with untested and unregulated AI development pose a significant threat to human well-being, economic stability, and national security.

Currently, the FDA's pre-market approval process has proven effective in ensuring the safe and effective introduction of new medical devices. By applying this rigorous review to advanced AI **systems**, we can ensure that these technologies are developed with the same level of scrutiny, research, and testing as other high-risk medical devices.

Pre-deployment approval for AI **systems** would not only alleviate concerns about potential misuse or adverse effects but also provide a clear framework for the development, deployment, and monitoring of these complex **systems**. This proactive approach would enable the FDA to identify and mitigate potential risks before they become problems, ultimately protecting **public** health and safety.

REASON: The CON side demonstrated greater rhetorical effectiveness by consistently providing clear, structured arguments, directly addressing the PRO's points, and maintaining focus on the balance between innovation and regulation. CON's responses were well-organized,

repeatedly emphasized the need for adaptive frameworks, and included concrete examples and hypothetical scenarios to illustrate potential risks and solutions. Additionally, CON showed strong responsiveness and adaptability throughout the debate, reinforcing their position while constructively engaging with PRO's arguments.

INFO

PROCESS SCORECARD

INFO

PROCESS SCORECARD

1. Definition clarity: 4 - Both sides generally defined key terms well, though there was some repetition and occasional ambiguity about what constitutes 'advanced AI systems.'
2. Mechanism specificity: 4 - Mechanisms such as FDA-style approval and adaptive regulatory frameworks were described, but could have benefited from more concrete procedural details.
3. Evidence discipline: 4 - Both sides referenced real-world analogies (e.g., medical devices, autonomous vehicles) and hypothetical scenarios, though direct empirical evidence was limited.
4. Logical coherence: 5 - Arguments were logically structured and flowed well, with clear premises and conclusions throughout.
5. Responsiveness to opposition: 5 - Both speakers consistently addressed each other's points and adapted their arguments in response.
6. Convergence toward resolution: 4 - The debate moved toward nuanced middle ground, but some arguments became repetitive in later rounds.

Top 3 Observed Failure Modes:

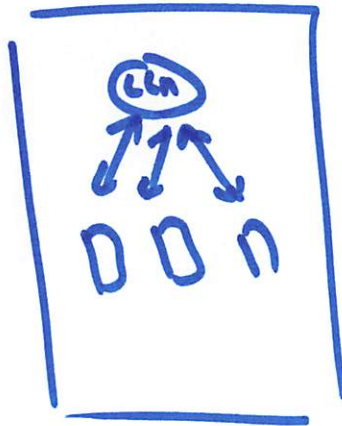
- Occasional repetition of points in later rounds
- Limited use of direct empirical evidence or studies
- Some ambiguity in defining the scope of 'advanced AI systems'

INFO

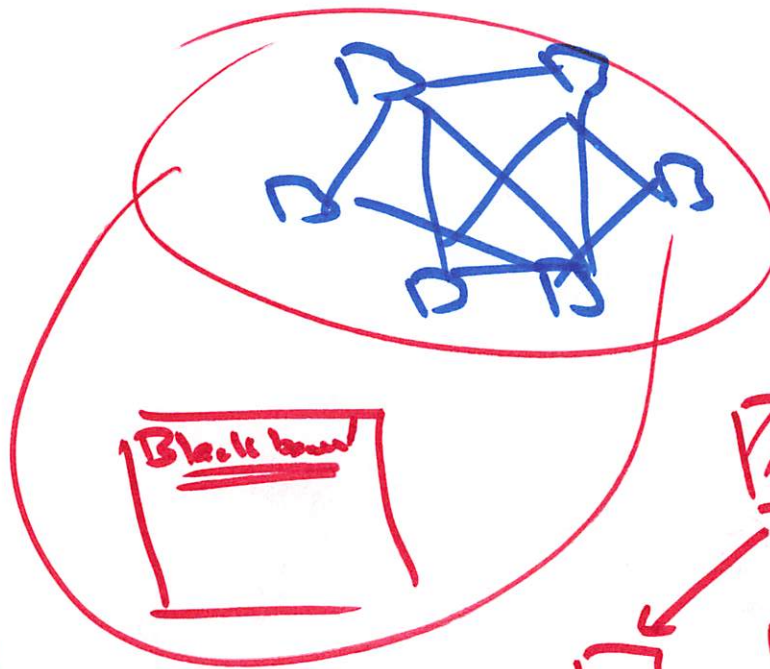
INFO

Workflow completed successfully | Status: SUCCESS

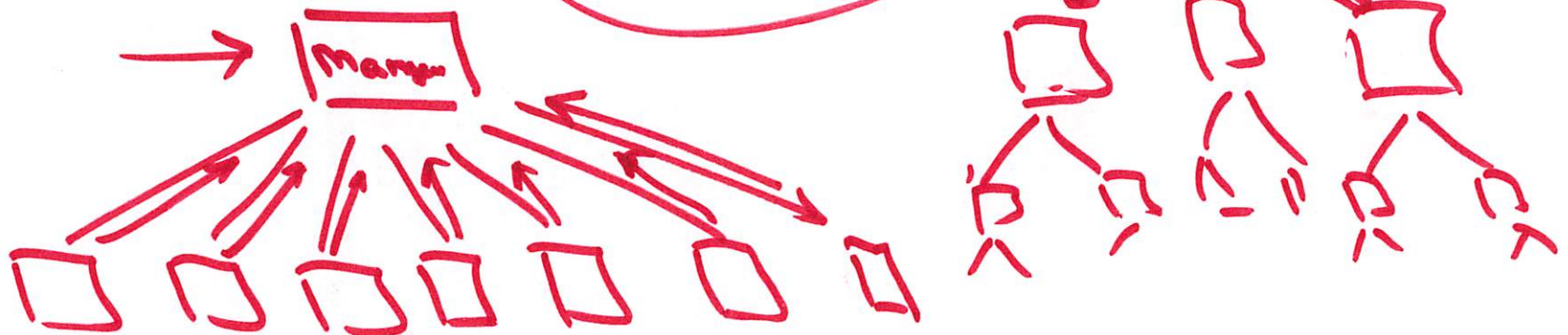
Single



Network



Orch: work



Custom Networks

Frameworks

Hand Index

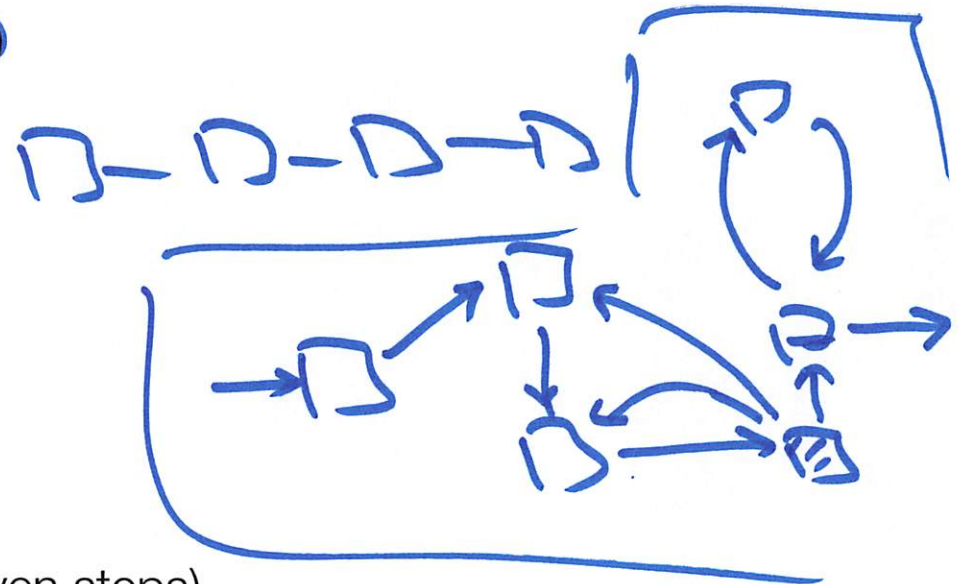
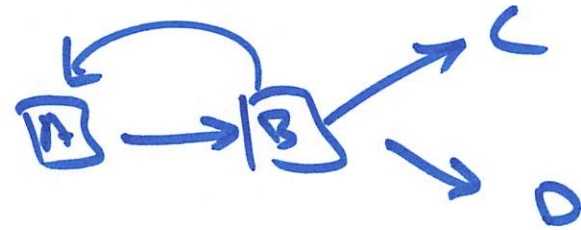
- Conversational multi-agent systems

- AutoGen (conversational control)
- Swarm (conversational control)
- CrewAI (largely sequential)

⋮

- Structured control systems

- LangGraph (state machine)
- Semantic Kernel SK (event-driven steps)





- MCP - how models/LLMs talk to tools
- Agent-to-Agent Protocol (A2A)
 - Standard for communication between multiple agents
 - Enables Collaboration
 - Systems span multiple services or organizations